# SUPPLY CHAIN ANALYTICS

Interactive report converted to PDF | Based on the supplied Supply_Chain.ipynb notebook

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### Key findings at a glance

| Orders | Late delivery rate | On-time rate | 90th percentile delay |
|---|---|---|---|
| **13,920** | **54.25%** | **45.75%** | **3 days** |
| Positive order profit sum | Profit sum for delayed records | Model accuracy | Class 1 recall |
| **$603.8K** | **$151.5K** | **69.00%** | **56.71%** |

# 1. Executive summary

The notebook analyzes 13,920 order records and identifies a material delivery reliability issue: 54.25% are delayed under its derived delay flag. The reported positive order profit sums to $603.8K. The notebook's delayed-order profit sum is $151.5K; this is a sum of order profit values for delayed records, not a separately established causal loss.

• Reduce late shipments and prioritize operational investigation on the highest-delay regions, categories, and time windows.

• Validate metric definitions before setting operational targets.

# 2. Business context & objectives

Supply chain performance connects fulfillment speed, customer experience, and order economics. This notebook focuses on order delay, profitability, regional/category bottlenecks, temporal patterns, and a classification model for late-delivery risk.

• Measure delay incidence and timing relative to scheduled shipment days.

• Compare order profit by delay duration.

• Identify segments and regions with higher late-delivery rates.

• Explore month, weekday, and hour patterns.

• Evaluate a Random Forest model for late-delivery risk.

# 3. Key performance indicators

| KPI | Notebook result | Interpretation |
|---|---|---|
| Total orders | 13,920 | Analysis population |
| Late deliveries | 7,552 | Delay flag is true where calculated Delay > 0 |
| Late delivery rate | 54.25% | More than half of records delayed |
| On-time delivery rate | 45.75% | Complement of late delivery rate |
| 90th percentile delay | 3 days | Among delayed records |
| Positive order profit sum | $603.8K | Sum where Order Profit Per Order > 0 |
| Profit sum for delayed records | $151.5K | Notebook calculation; not causal loss attribution |

# 4. Profitability analysis

Profitability labels in the notebook show approximately 80.68% profitable, 18.73% loss, and 0.59% break-even records. Mean profit declines for delays of 2–4 days versus early/on-schedule orders, although total profit by delay is affected by order volume.

Delay is calculated as actual shipping date minus order date (in days), less scheduled shipment days. Values below zero indicate shipment earlier than scheduled; zero indicates on schedule; positive values indicate days late. Aggregate profit is not evidence that delay caused a profit change.

• Use both total profit and mean profit when comparing delay duration.

# 5. Bottleneck detection

The notebook calculates late-delivery percentage by Order Region, Customer Segment, Order Status, Type, Market, and Department Name. These comparisons are intended to surface groups with elevated delay rates.

Category-level values are not available as structured output in the saved notebook. A ranked bottleneck list should be generated by re-running the notebook and exporting grouped tables.

# 6. Root cause analysis

A regional driver function evaluates Customer Segment, Department Name, Type, and Order Status within a selected region; the notebook runs it for East Africa.

• These are descriptive associations, not confirmed root causes.

• Shipping Mode is absent from the analysis after cleaning, so it cannot be evaluated here.

• Operational causes such as carrier capacity, inventory availability, warehouse processing, or route conditions require additional data and validation.

# 7. Time based daily patterns

The notebook analyzes delay percentage by month, weekday, and order hour. The saved output includes hourly values; the highest observed hour in that table is 20:00 at 61.90%.

Hourly and calendar patterns are descriptive. Validate sample sizes and investigate whether timestamps represent comparable local operating time zones before changing staffing or cutoffs.

# 8. Machine learning model results

The notebook trains a Random Forest classifier to predict Late_delivery_risk using Type, scheduled shipment days, Category Name, Customer Segment, Department Name, Order Region, Order month, and Order hour. Categorical columns are frequency-encoded; training data is balanced with SMOTE.

| Metric | Result | Interpretation |
|---|---|---|
| Accuracy | 69.00% | Overall test accuracy |
| Class 1 precision | 78.89% | Among predicted risk cases, share truly in class 1 |
| Class 1 recall | 56.71% | Model identifies just over half of actual class 1 records |
| Test records | 2,784 | Held-out test set |
| Class 0 metrics | P 0.63 / R 0.83 / F1 0.72 | Support: 1,308 |
| Class 1 metrics | P 0.79 / R 0.57 / F1 0.66 | Support: 1,476 |

**Class 1 recall of 56.7% means the model misses a substantial share of actual risk-class records. This is a baseline, not production-ready evidence.**

# 9. Strategic recommendations

• Establish a reliable delivery dashboard: track late rate, delay days, and on-time rate by region, market, department, and segment.

• Investigate high-delay groups: use grouped tables to prioritize specific bottlenecks, then validate with operational owners.

• Separate correlation from cause: enrich with carrier, warehouse, inventory, route, and capacity data before attributing delays.

• Review promise-date planning: compare scheduled days against actual distribution and revisit service-level assumptions.

• Improve model evaluation: test temporal validation, prevent target leakage, report confusion matrix/ROC-AUC and calibration, and tune recall/precision to business costs.

• Monitor impact: pilot interventions and measure late rate and customer/business outcomes against a baseline.

## 10. Conclusion

The notebook indicates delivery performance is a significant area for improvement, with a late rate above 54%. Profitability and delay patterns vary across delay durations, and the baseline Random Forest has moderate predictive performance. The next step is to make category/region findings actionable by exporting ranked data, validating causes with operational evidence, and measuring improvement through a recurring KPI dashboard.

Source: user-provided Supply_Chain.ipynb. Interpretive limits are noted where the notebook does not establish causal claims.